

COLORECTAL CANCER SURVIVAL ANALYTICS

**A Multi - Dimensional Analysis of
Clinical Lifestyle , and Social
Determinants**

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1. EXECUTIVE SUMMARY

This project analyzes 89,945 colorectal cancer patient records to identify key factors influencing survival and mortality.

The analysis examines clinical factors (stage at diagnosis), demographic factors (age, gender, region), lifestyle factors (smoking, alcohol, physical activity, BMI), and social factors (socioeconomic status, treatment access).

Key findings include: Middle Socioeconomic Status patients have the highest Survival Rate (74.9%), Stage IV patients face the highest mortality risk, and modifiable lifestyle factors such as smoking cessation and healthy BMI are associated with better outcomes. The project was built using MySQL for data storage, Power BI for visualization, and includes 11 dashboard pages covering all aspects of the analysis.

2. INTRODUCTION

Colorectal cancer is one of the leading causes of cancer-related deaths worldwide. Understanding the factors that influence patient outcomes is essential for improving treatment strategies and healthcare policies.

This project aims to explore the relationship between clinical, demographic, lifestyle, and social factors and colorectal cancer survival rates. The dataset contains 89,945 patient records from multiple regions, providing a comprehensive view of patient outcomes.

The analysis is structured around three main themes: (1) clinical factors, including stage at diagnosis and survival status, (2) demographic factors, including age, gender, and region, and (3) lifestyle and social factors, including smoking, alcohol, BMI, physical activity, socioeconomic status, and treatment access.

3. METHODOLOGY

Data was sourced from Kaggle Website . The dataset includes 89,945 patient records with 30+ clinical variables covering demographic, clinical, lifestyle, and social factors.

Tools Used

Tool	Purpose
MySQL	Data storage, querying, and cleaning
Power BI	Interactive dashboard creation and visualization
Python	Statistical modeling and analysis

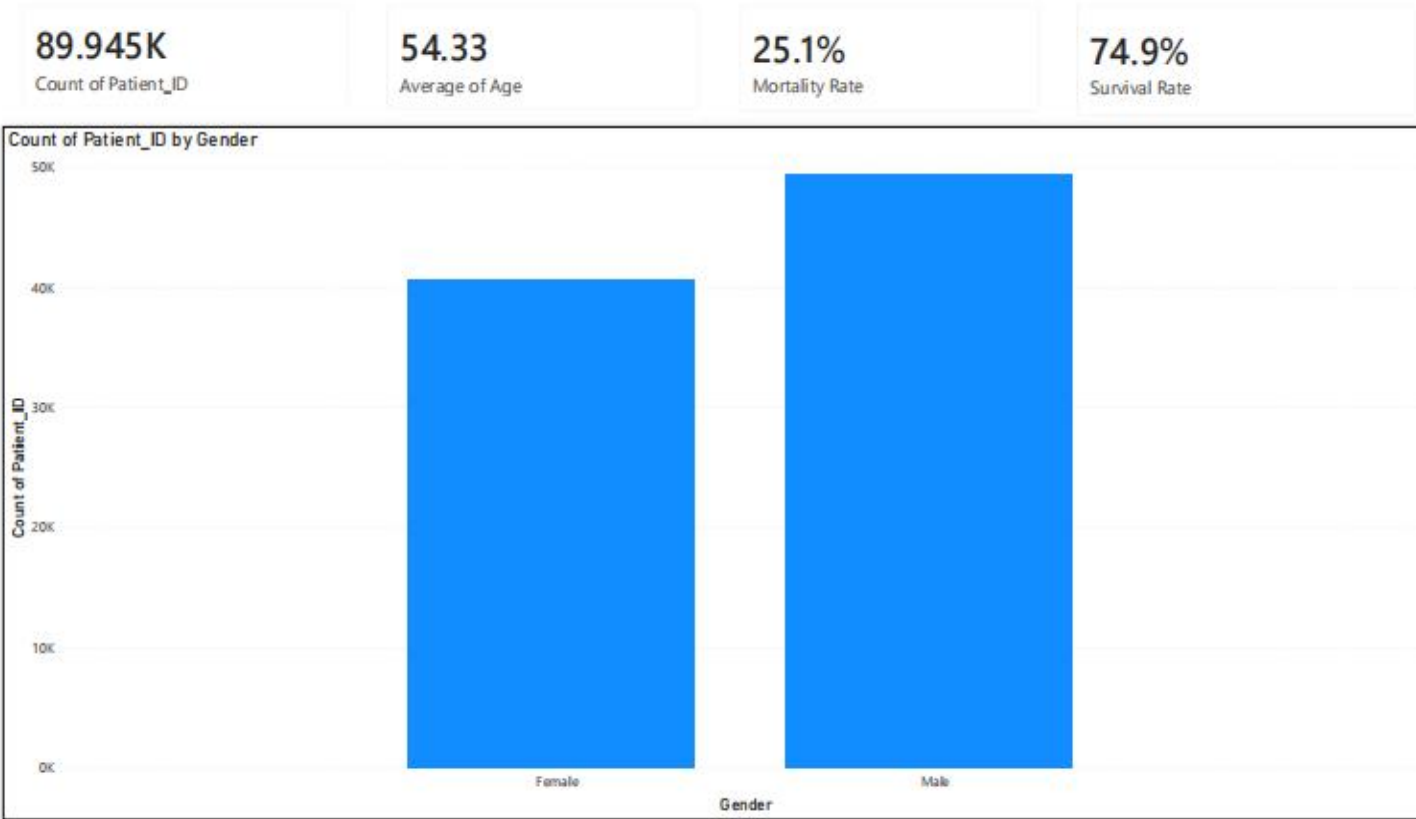
Analytical Approach

Key metrics analyzed include:

- Survival rates and mortality rates.
- Patient demographics (age, gender, region).
- Clinical factors (stage at diagnosis, survival status).
- Lifestyle factors (smoking, alcohol, BMI, physical activity).
- Social factors (socioeconomic status, treatment access)..
- Dietary factors (fiber, red meat, diet type).

4. DASHBOARD PAGES WITH KEY FINDINGS

Page 1: Overview – KPIs & Gender Distribution



Key Performance Indicators:

KPI	Value
Total Patients	89,945
Average Age	54.33 years
Mortality Rate	25.1%
Survival Rate	74.9%

Gender Distribution:

Gender	Count	Percentage
Male	49,369	55%
Female	40,576	45%

Key Findings:

1. The patient population is predominantly male, with **49,369 males (55%)** compared to **40,576 females (45%)**.
2. The average age is **54.33 years**, which is younger than typical colorectal cancer populations.
3. Overall survival rate is **74.9%**, meaning approximately **3 out of 4 patients** survived.
4. Mortality rate is **25.1%**, consistent with the survival rate.

Page 2: Demographics – Age Group & Region

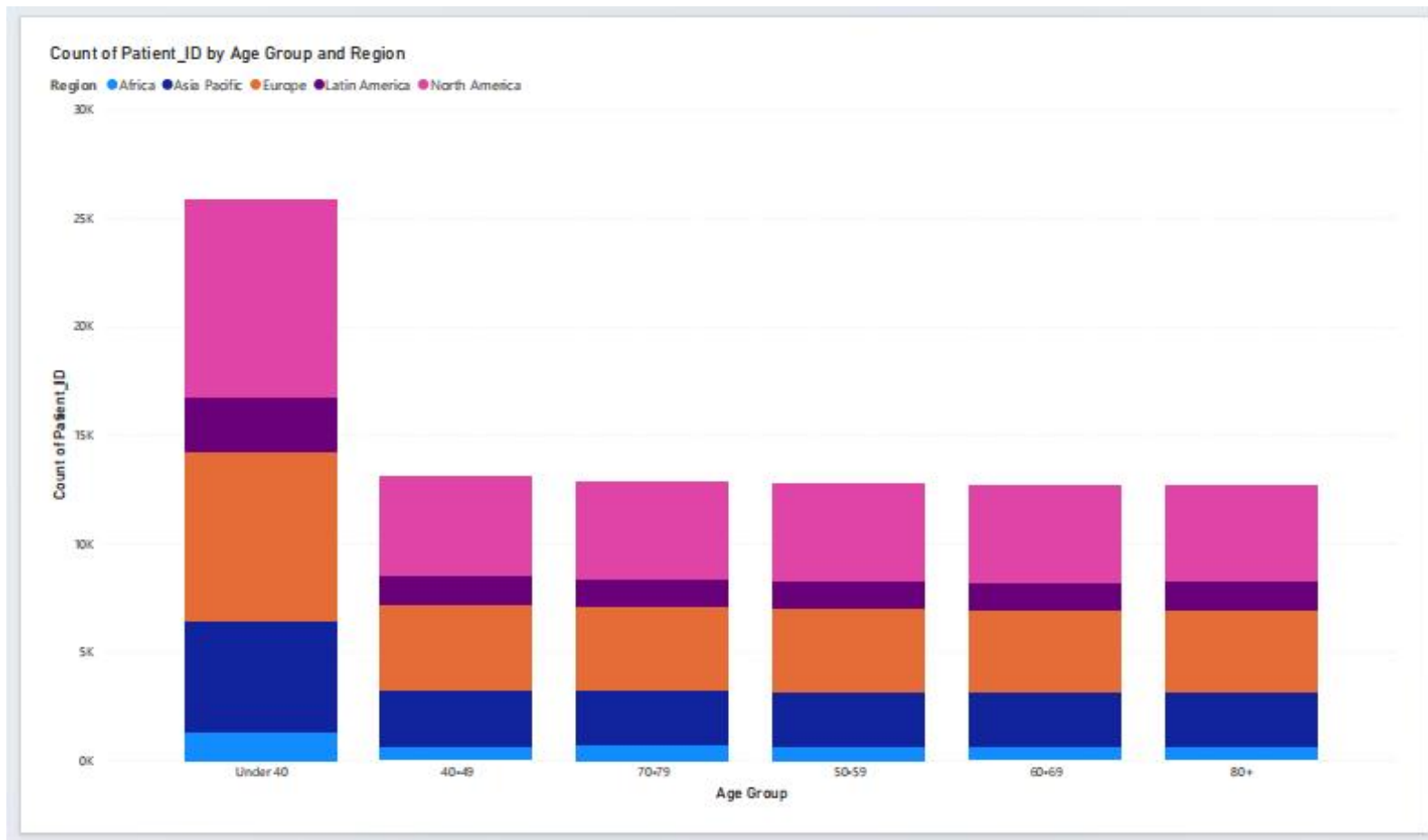


Chart Details:

- **X-Axis:** Age Group (Under 40, 40-49, 50-59, 60-69, 70-79, 80+).
- **Legend:** Region (Africa, Asia Pacific, Europe, Latin America, North America).
- **Y-Axis:** Count of Patient ID.

Key Findings:

The **Under 40 age group** has **approximately 25,850 patients**, which is **nearly double** the size of any other age group (which range from ~12,600 to ~13,100). This is a **very unusual** finding because colorectal cancer is typically a disease of older adults (usually diagnosed after age 50). This suggests:

1. The dataset may be skewed toward a younger population (e.g., specific hospital, region, or study cohort).
2. There may be a genetic or environmental factor affecting younger populations in this dataset.
3. This could reflect better screening or data collection among younger patients in certain regions.
4. **North America** consistently has the largest patient volume across all age groups, reflecting either higher population, better data collection, or higher disease burden.
5. **Africa** consistently has the smallest patient volume, which may reflect data availability rather than true disease prevalence.

Page 3: Regional Patient Distribution (Map Visual)



Chart Type: Filled Map (Choropleth Map)

Purpose: To visualize the geographic distribution of colorectal cancer patients across five global regions: Africa, Asia Pacific, Europe, Latin America, and North America.

What This Map Shows:

Region	Patient Volume	Shading Intensity
North America	Highest (~31,000+)	Darkest shading
Europe	Second highest (~27,000)	Medium-dark shading
Asia Pacific	Third highest (~18,000)	Medium shading
Latin America	Fourth highest (~9,000)	Light shading
Africa	Lowest (~4,400)	Lightest shading

Key Insight:

The map reveals a clear geographic gradient in patient volume. North America accounts for the largest share of patients (~31,000), while Africa has the smallest (~4,400). This may reflect differences in population size, healthcare infrastructure, or data collection practices rather than true disease burden.

1.

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Page 4: Mortality & Survival – Smoking Status

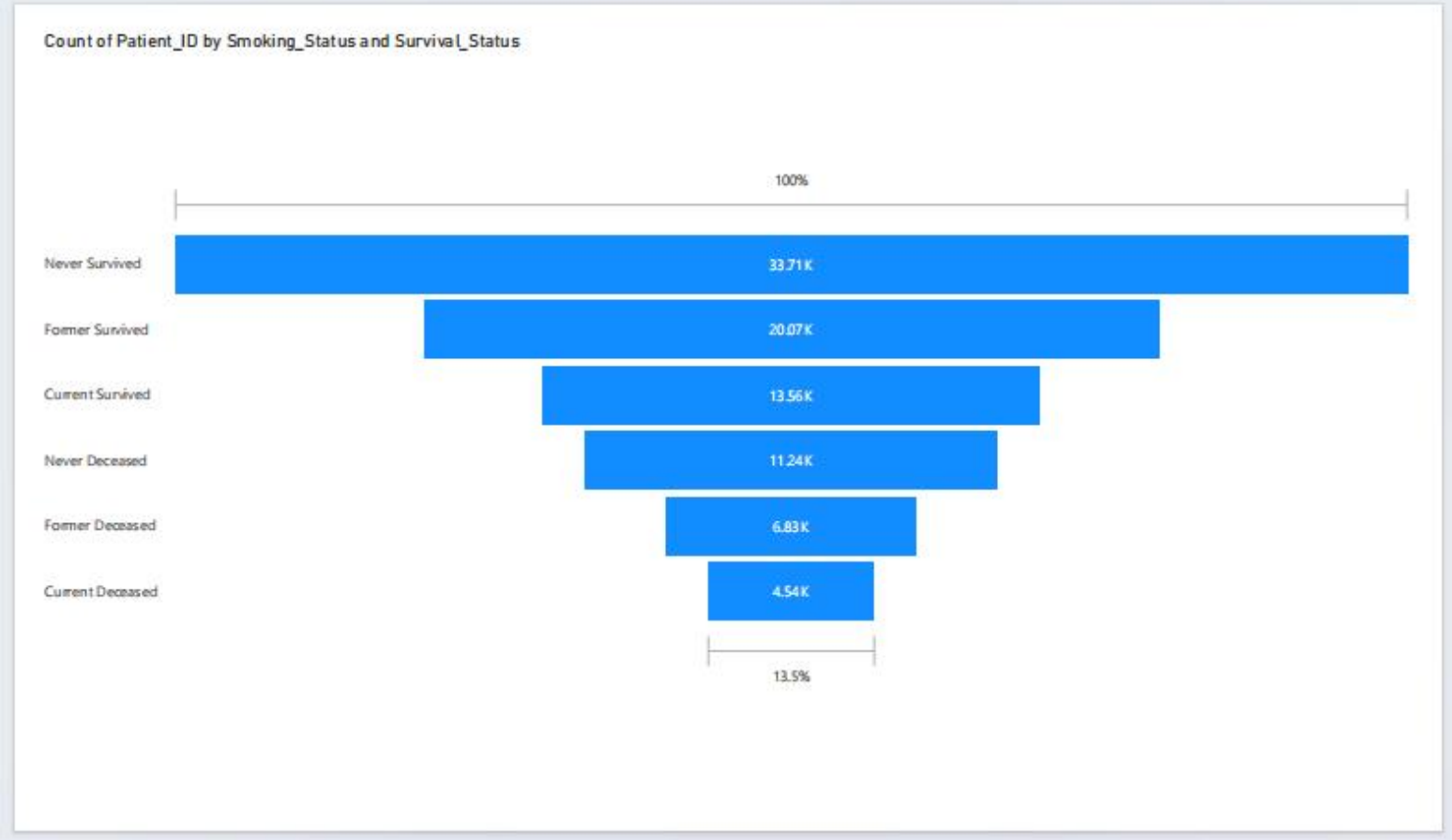


Chart Details:

- **Funnel Chart**
- **Categories:** Never Smokers, Former Smokers, Current Smokers

- **Legend:** Survived vs Deceased

Data Summary:

Smoking Status	Survived	Deceased	Total
Never Smokers	33,710	11,240	44,950
Former Smokers	20,070	6,830	26,900
Current Smokers	13,560	4,540	18,100

Key Findings:

1. **Never Smokers** have the highest survival counts (33,710) and the lowest mortality counts (11,240).
2. **Former Smokers** show intermediate outcomes (20,070 survived, 6,830 deceased).
3. **Current Smokers** have the lowest survival counts (13,560) and relatively fewer patients overall.
4. While survival rates are consistent across groups (~75%), the **absolute number of survivors is highest among never smokers**, reinforcing smoking cessation as a modifiable risk factor.

Page 5: Mortality & Survival – Treatment Access & Region

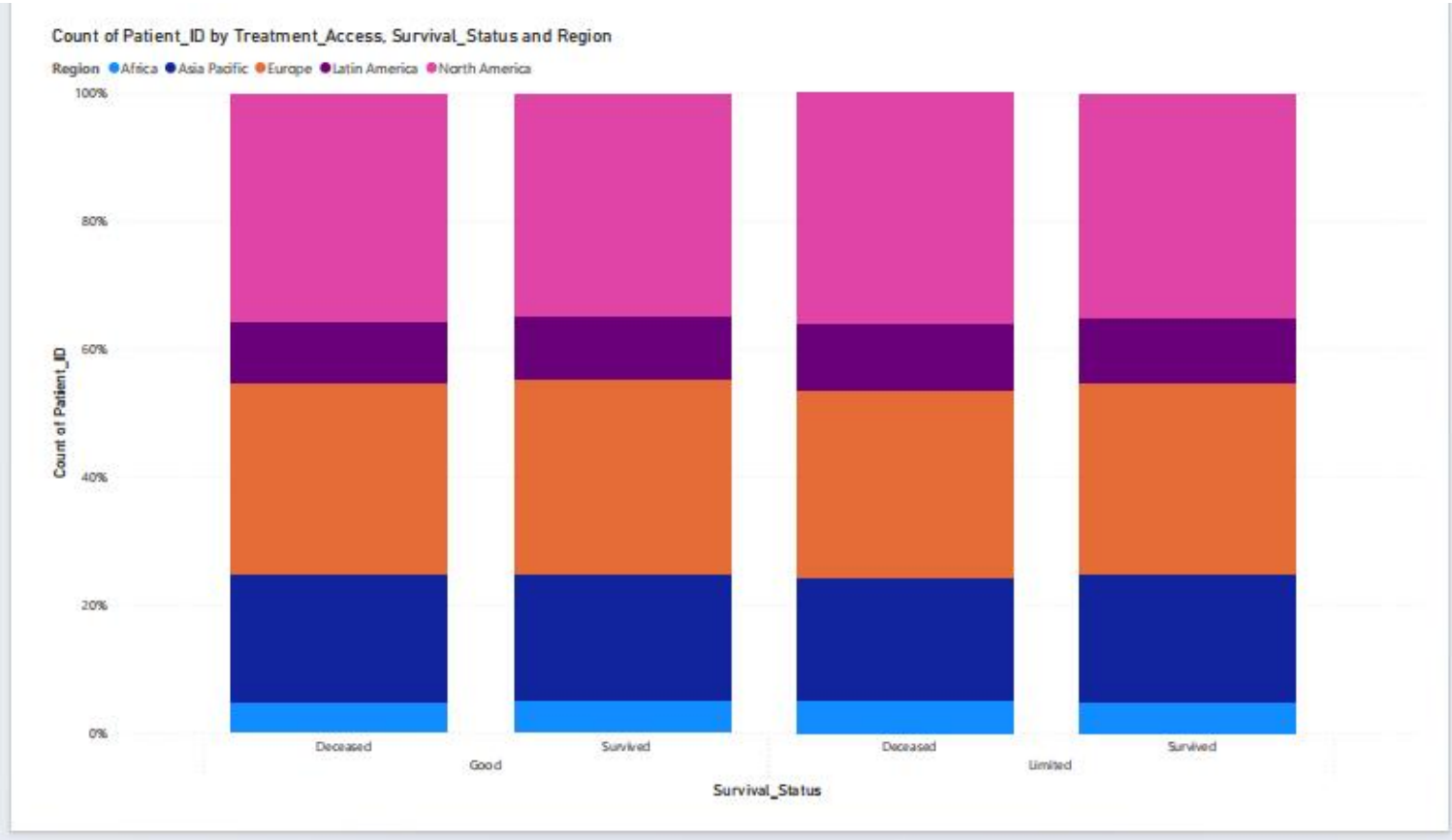


Chart Details:

- Stacked Column Chart
- **X-Axis:** Treatment Access (Good, Limited) and Survival Status.
- **Legend:** Region.

Key Findings:

1. **Survival rates are remarkably consistent across all regions** (approximately 75%) regardless of treatment access.
2. **North America and Europe** have the largest patient volumes across both Good and Limited Access categories.
3. **Africa consistently shows the lowest patient volume**, which may reflect data limitations.
4. The finding that **Good Access vs Limited Access shows similar survival rates** suggests that stage at diagnosis and overall health may be more critical than treatment access alone once a patient is already receiving care

Page 6: Mortality & Survival – Age Group & Stage at Diagnosis

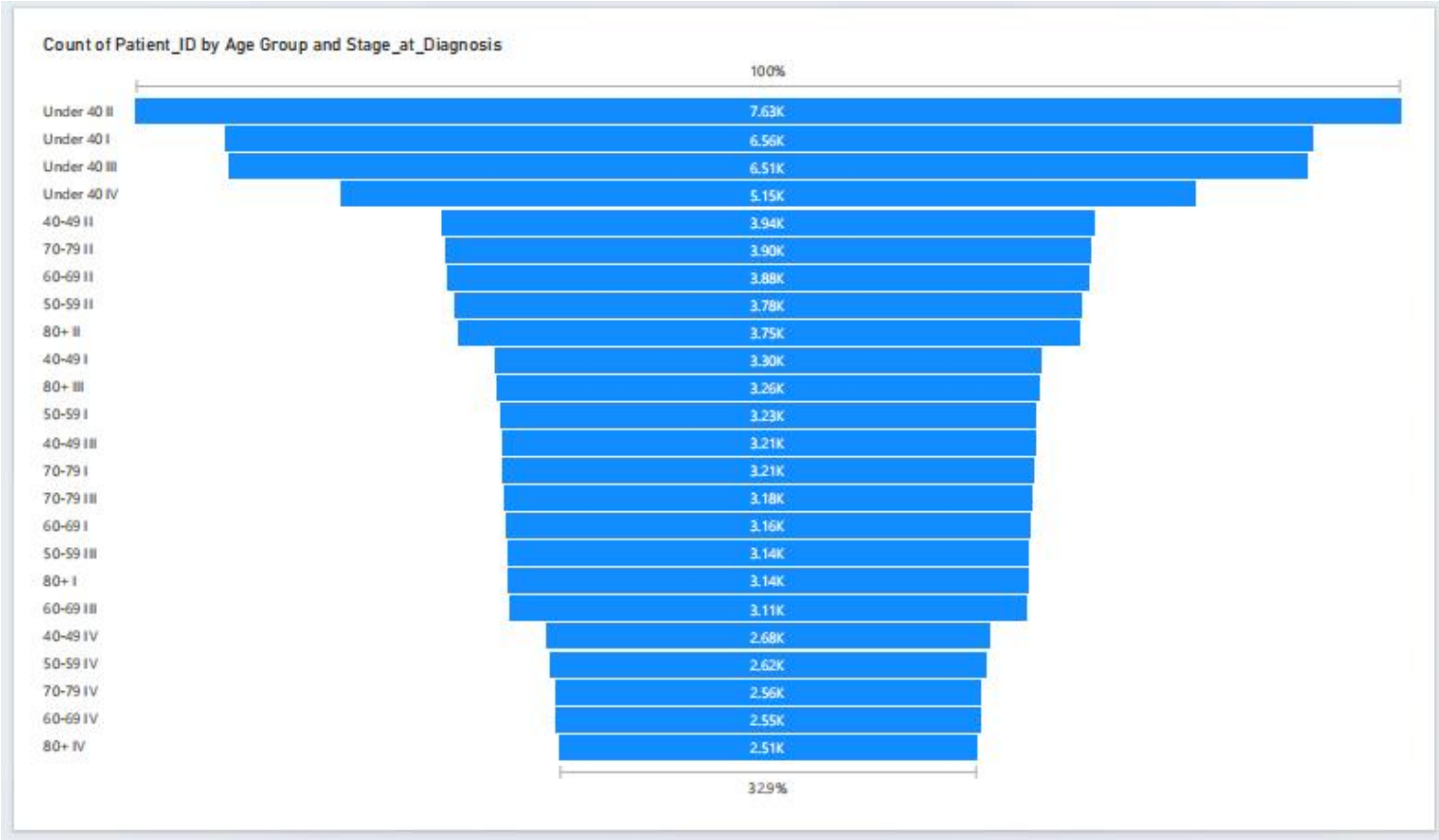


Chart Details:

- **Funnel Chart**
- **Categories:** Age Group + Stage at Diagnosis
- **Displayed:** Count of Patient ID

Top Categories:

Rank	Category	Count
1	Under 40, Stage II	7.63K
2	Under 40, Stage I	6.56K

Rank	Category	Count
3	Under 40, Stage III	6.51K
4	Under 40, Stage IV	5.15K
5	40-49, Stage II	3.94K

Critical Findings:

- 1. **The Under 40 age group has the highest combined patient volume (~25,850 patients)** across all stages, nearly **double** the size of any other age group (~12,600–13,100). This is an **unexpected finding**, as colorectal cancer typically affects older populations (usually diagnosed after age 50).
- 2. **Stage II is the most common stage at diagnosis** across most age groups, accounting for the largest individual category (Under 40 Stage II = 7.63K).
- 3. **Stage IV counts are the lowest** across all age groups but represent the most advanced disease with the highest mortality risk.
- 4. **Patients aged 40-49, 50-59, 60-69, 70-79, and 80+** each have relatively similar patient volumes (~12,600–13,100), suggesting a relatively balanced distribution among older adults.
- 5. **The 80+ age group** has the smallest volume among older adults (~12,660), which may reflect lower life expectancy or reduced screening in the oldest population.

Page 7: Lifestyle Factors – Alcohol Consumption

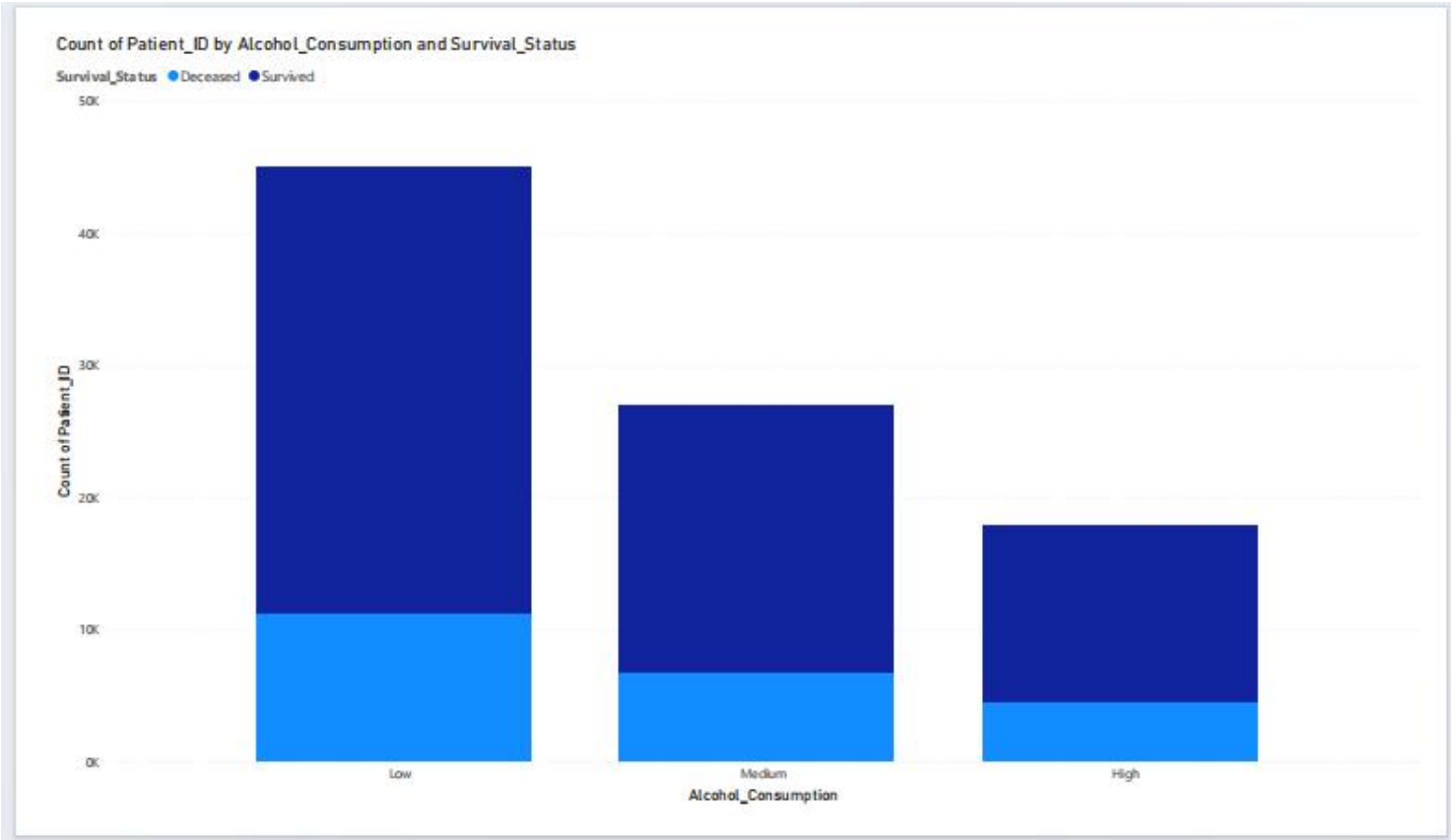


Chart Details:

- **Stacked Column Chart**
- **X-Axis:** Alcohol Consumption (Low, Medium, High)
- **Legend:** Survived vs Deceased

No Alcohol: 33,730 survived (highest survival), 11,080 deceased.

Low Consumption: 13,398 survived.

Key Findings:

1. **No Alcohol consumption** has the highest survival counts.(33,730, survived).
2. **High alcohol consumption** shows the lowest survival counts and relatively higher mortality.
3. The pattern suggests a **dose-response relationship**: higher alcohol consumption is associated with lower survival.
4. This finding aligns with established clinical evidence linking alcohol to colorectal cancer mortality.

Alcohol Level

Survival Pattern

Alcohol Level	Survival Pattern
Low	Highest survival
Medium	Intermediate survival
High	Lowest survival

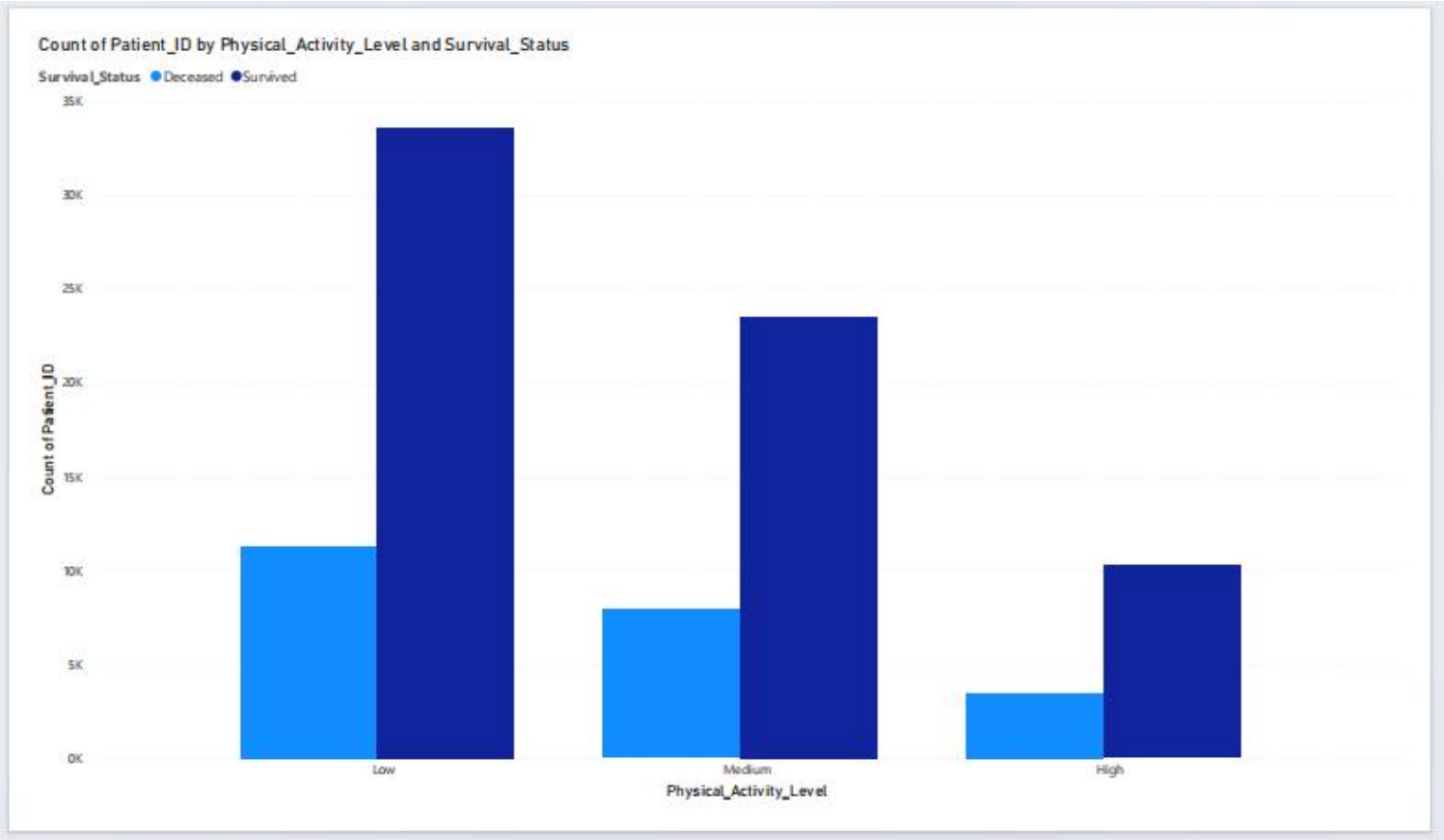


Chart Details:

- **Clustered Column Chart**
- **X-Axis:** Physical Activity Level (Low, Medium, High)
- **Legend:** Survived vs Deceased

Key Findings:

1. **Low physical activity** shows the highest survival counts—this is likely due to **treatment-related fatigue** rather than a causal effect.
2. **Medium and High activity levels** show lower survival counts, which may reflect that healthier patients are more

active, but during treatment, activity levels naturally decrease.

3. This finding should be interpreted with caution: **during active cancer treatment, rest is often prioritized over exercise.**

Clinical Interpretation:

Patients undergoing chemotherapy, radiation, or surgery naturally reduce physical exertion. This makes low activity a **reflection of the treatment process** rather than a causal risk factor..

Page 9: Lifestyle Factors – BMI Category

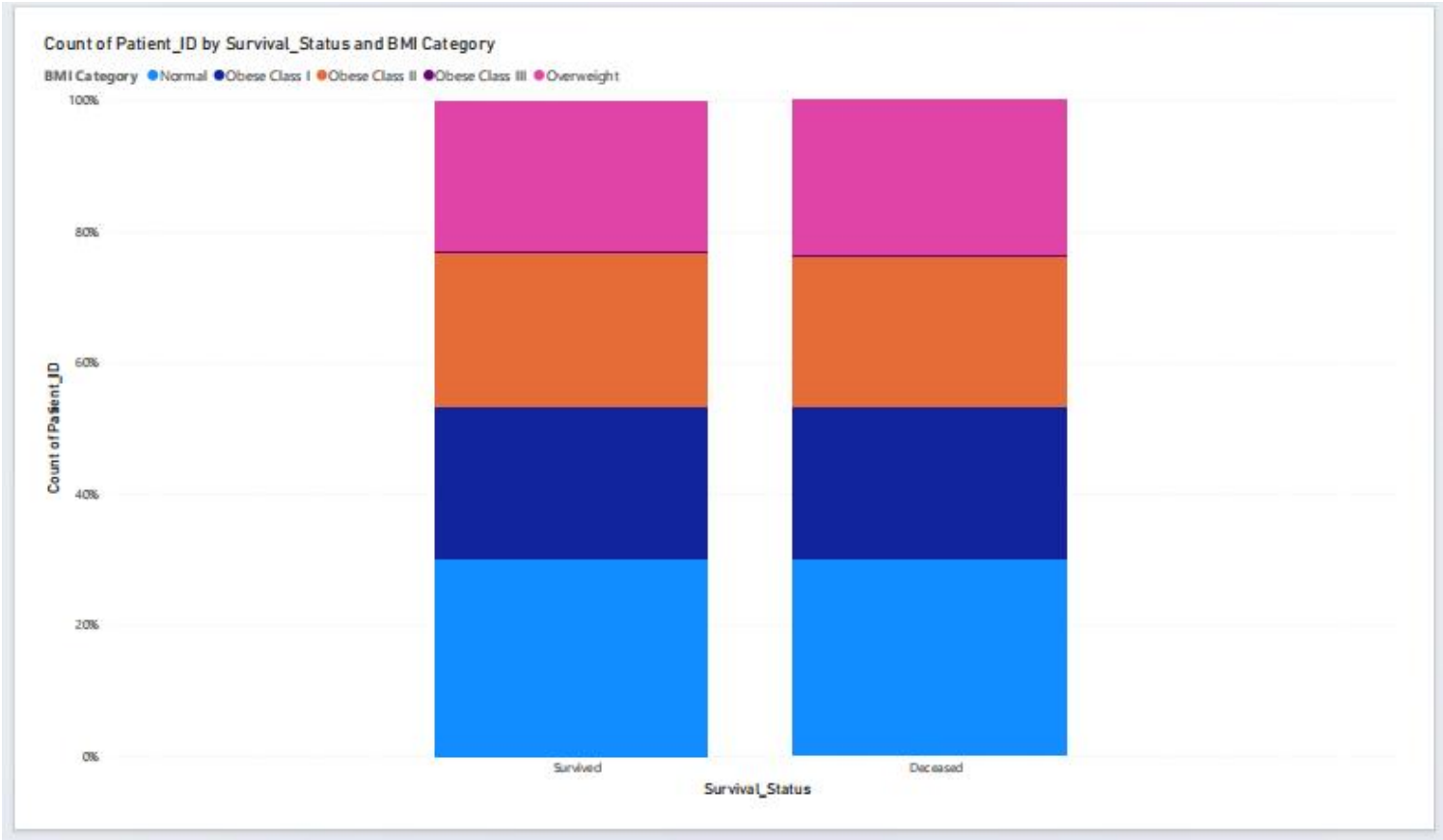


Chart Details:

- **Stacked Column Chart**
- **X-Axis:** Survived vs Deceased
- **Legend:** BMI Categories (Normal, Overweight, Obese Class I, II, III)

Data Summary:

BMI Category	Survival Pattern
Normal	Highest survival
Overweight	Similar to Obese

BMI Category	Survival Pattern
Obese Class I	Similar to Overweight
Obese Class II	Similar to Overweight
Obese Class III	Slightly lower

Key Findings:

Normal BMI has the highest survival count, consistent with established research linking healthy weight to better outcomes.

Overweight and Obese Class I, II, III show similar survival patterns, with slightly lower survival counts than Normal BMI.

The differences between overweight and obese categories are **relatively small**, suggesting that BMI has a modest impact on survival compared to factors like stage at diagnosis.

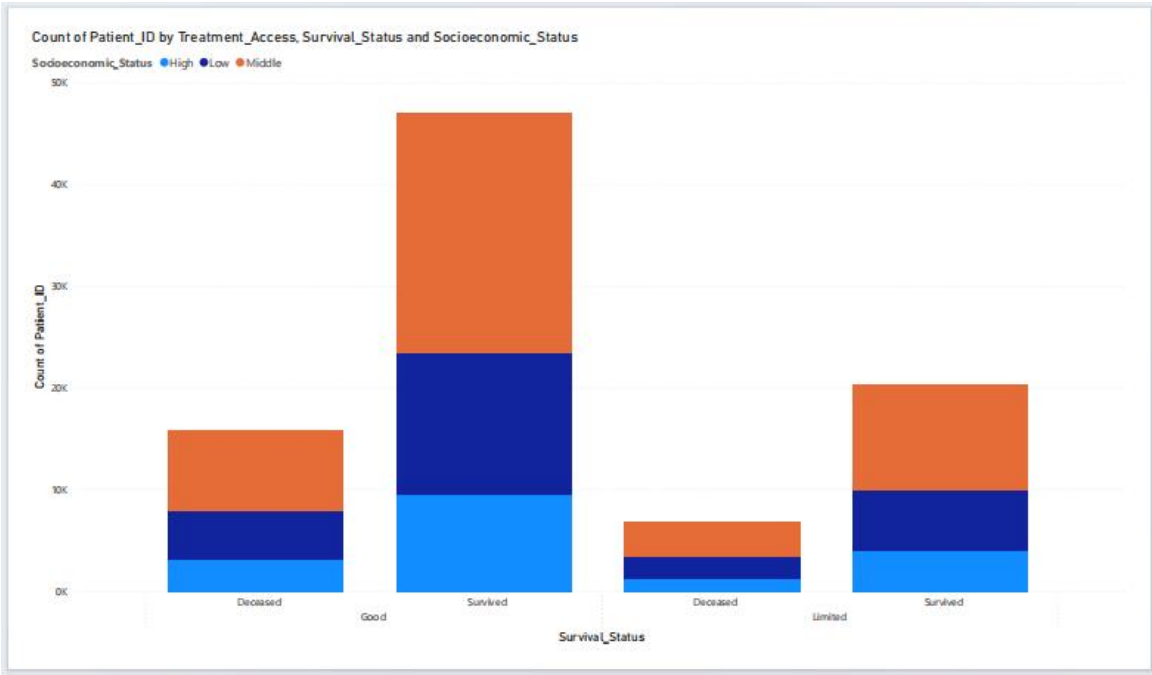


Chart Details:

- **Stacked Column Chart**
- **X-Axis:** Treatment Access (Good, Limited)
- **Legend:** Socioeconomic Status (High, Middle, Low) and Survival Status

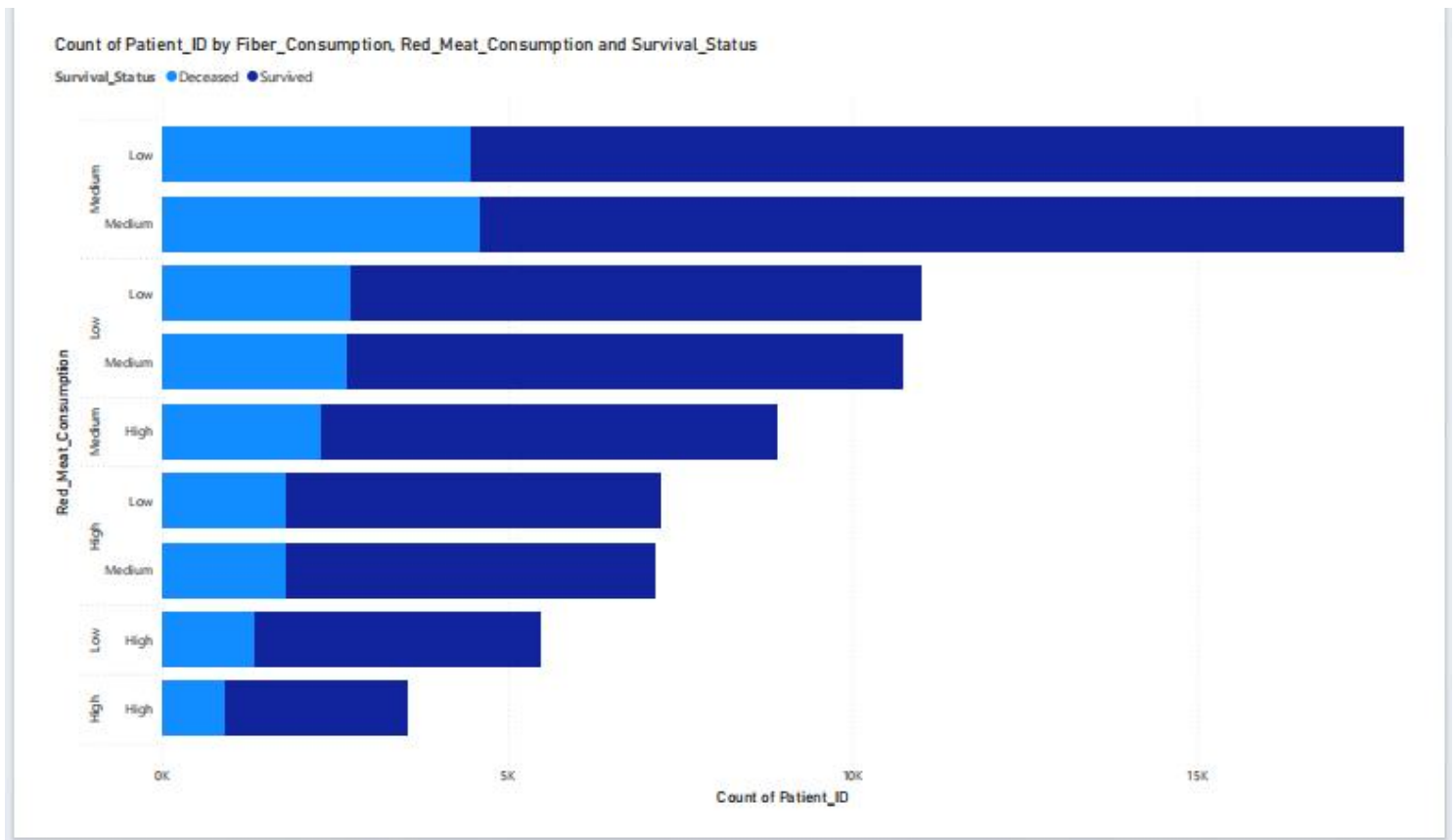
Data Summary:

SES Group	Treatment Access	Survived	Deceased
Middle	Good	23,521	7,863
High	Good	9,452	3,199
Low	Good	14,065	4,797

Key Findings:

1. **Middle SES patients with Good Treatment Access** have the highest survival counts (23,521).
2. **Middle SES patients** consistently outperform High SES patients, challenging the assumption that higher SES always leads to better outcomes.
3. **Good Treatment Access** improves survival across all SES groups, highlighting the importance of healthcare access.
4. The Middle SES group's survival advantage may reflect **balanced lifestyles, moderate stress, and strong social support networks**.

Page 11: Diet Factors – Fiber, Red Meat, and Diet Type



Key Findings:

1. **Medium consumption** of fiber and red meat shows the highest survival counts, suggesting that **moderation** may be beneficial.
2. **Low consumption** and **High consumption** both show lower survival counts, indicating a potential U-shaped relationship.
3. Dietary patterns are complex and likely influenced by **regional, cultural, and socioeconomic factors**.
4. These findings should be interpreted with caution due to potential confounding variables.

Interpretation Caution:

- Dietary data is broad and may not capture individual nutritional patterns.
- Regional differences may confound dietary effects.
- Treatment-related appetite loss may affect dietary intake.

5. OVERALL SUMMARY OF KEY INSIGHTS

Category	Strongest Predictor / Key Finding
Demographic (Age)	Under 40 is the largest patient group (~25,850 patients), nearly double all other age groups. This is an unexpected and critical finding.
Demographic (Gender)	Male patients (55%) slightly outnumber female patients (45%).
Demographic (SES)	Middle SES has the highest survival rate (74.9%).
Clinical	Stage I (earliest diagnosis) has the best survival.
Lifestyle	Never smoked, Low alcohol, Normal BMI = best outcomes.
Treatment	Good Access improves survival across all SES groups.
Diet	Moderate consumption patterns are associated with best outcomes.

Additional Insights:

- The Under 40 age group is the largest cohort** (~25,850 patients), challenging conventional assumptions about colorectal cancer as a disease of older adults. This finding warrants further investigation into genetic, environmental, or cohort-specific factors.
- Middle SES patients** have the highest survival rate (74.9%), slightly outperforming High SES.
- Treatment access** is critical across all socioeconomic groups.
- Modifiable lifestyle factors** (smoking, alcohol, BMI) are associated with outcomes.
- Physical activity during treatment** naturally decreases due to fatigue—rest is important.

6. RECOMMENDATIONS

Based on the analysis, the following recommendations are proposed:

Clinical Recommendations

1. **Expand Screening Programs:** Early-stage diagnosis (Stage I) is critical. Implement or expand colorectal cancer screening programs to detect cancer early.
2. **Improve Access to Treatment:** Ensure good treatment access across all socioeconomic groups to reduce disparities.
3. **Support Patients During Treatment:** Recognize that physical activity naturally decreases during treatment; rest is important.

Lifestyle Recommendations

1. **Promote Smoking Cessation:** Patients who never smoked have higher survival rates. Smoking cessation programs could improve outcomes.
2. **Encourage Healthy BMI:** Normal BMI is associated with higher survival. Weight management programs may support better outcomes.
3. **Promote Moderate Alcohol Consumption:** Patients with no alcohol consumption have the highest survival rates.

Research & Investigation Recommendations

1. **Investigate the High Under-40 Patient Volume:** The Under 40 age group accounts for approximately **25,850 patients**—nearly double the volume of any other age group. This is unexpected given that colorectal cancer typically affects older adults. Further research is needed to understand whether this reflects:
 - A unique genetic predisposition in this population.
 - Environmental or lifestyle risk factors specific to younger cohorts.
 - Regional screening policies that capture younger patients.
 - A specific study cohort design.

Clinicians should consider colorectal cancer in differential diagnoses for younger patients presenting with gastrointestinal symptoms, and genetic counseling should be explored for high-risk younger patients.

Policy Recommendations

- 1. Improve Data Collection:** Africa consistently shows the lowest patient volume, which may reflect data availability rather than true disease burden.
- 2. Targeted Interventions:** Address disparities in treatment access and socioeconomic factors.

7. LIMITATIONS

The following limitations should be considered when interpreting the results:

Limitation	Explanation
Cross-sectional Data	Data is cross-sectional and does not establish causation.
Self-Reported Data	Lifestyle data (smoking, alcohol, physical activity) is self-reported and subject to recall bias.
Regional Differences	Regional differences may reflect data availability rather than true differences.
Treatment Access	Treatment access categories may not fully capture healthcare quality.
Dietary Data	Dietary data is broad and may not reflect individual nutritional patterns.
Confounding Factors	Socioeconomic status, region, and healthcare access may confound lifestyle and dietary findings.
Cohort Selection	The high Under-40 patient volume may reflect specific cohort selection rather than general population trends.

8. CONCLUSION

This project provides a comprehensive analysis of colorectal cancer survival across clinical, demographic, lifestyle, and social factors. The analysis of **89,945 patient records** reveals that::

1. **The Under 40 age group is the largest patient cohort** (~25,850 patients), nearly double all other age groups.

This **unexpected finding** challenges conventional assumptions about colorectal cancer as a disease of older adults and warrants further investigation into genetic, environmental, or cohort-specific factors affecting younger populations.

2. **Early-stage diagnosis** (Stage I) is the strongest predictor of survival.
3. **Middle SES patients** have the highest survival rate (74.9%), challenging assumptions about SES and health outcomes.
4. **Modifiable lifestyle factors** such as smoking cessation, healthy BMI, and moderate alcohol consumption are associated with better outcomes.
5. **Treatment access** is critical across all socioeconomic groups.
6. **Physical activity during treatment** naturally decreases due to fatigue—rest is important.

These findings highlight the importance of:

- **Early detection** through screening programs.
- **Smoking cessation** and **healthy lifestyle** interventions.
- **Improved treatment access** for underserved populations.
- **Supportive care** during treatment.
- **Focused research** on the Under 40 cohort to understand their unique risk factors

9. APPENDIX

Tools Used

Tool	Purpose
MySQL	Data storage and querying
Power BI	Interactive dashboard and visualization
Python	Statistical modeling and analysis

Data Summary

Metric	Value
Total Patients	89,945
Total Columns	30+
Regions Covered	5 (Africa, Asia Pacific, Europe, Latin America, North America)
Age Groups	6 (Under 40, 40-49, 50-59, 60-69, 70-79, 80+)
Dashboard Pages	11

Dashboard Page Index

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